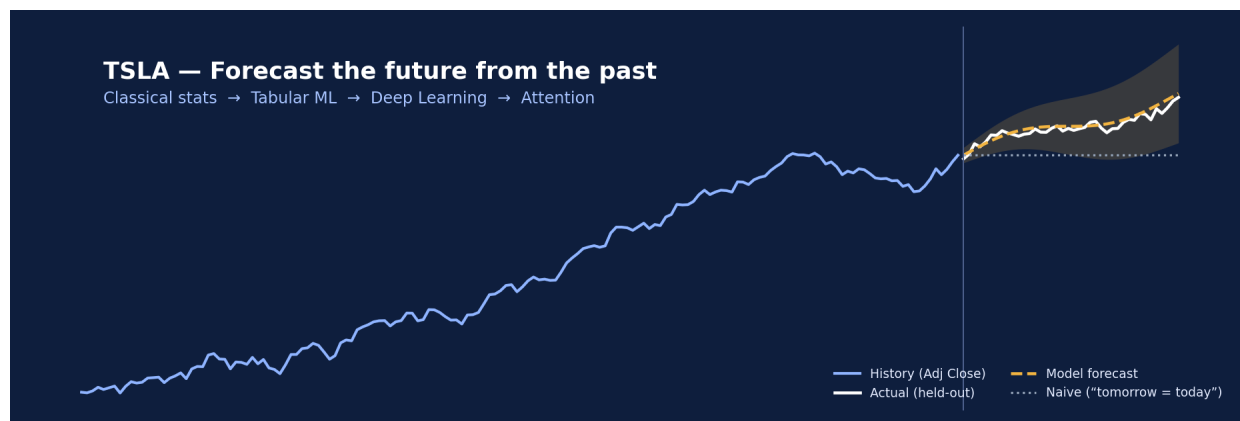


CP020003 — Artificial Intelligence 2026

Khon Kaen University

Week 9— Homework Assignment

In class this week you built a full forecasting ladder on the NASDAQ Composite index — classical statistical models (AR, MA, ARMA, ARIMA, SARIMA, Holt-Winters, VAR), tabular ML models fit on lag features (Linear Regression, Random Forest, XGBoost, LightGBM, CatBoost), deep sequence models (MLP, RNN, LSTM, GRU, 1D CNN), and a causal self-attention demo — all scored on the same held-out test window. This assignment is deliberately open-ended: you will use the exact same techniques, but on your own instrument (Tesla, TSLA), your own train/test split, and your own combination of model families to investigate a forecasting question you find genuinely interesting. There is no single “correct” model to land on — you are graded on how rigorously you split, forecast, diagnose, and compare, and how convincingly you turn the numbers into a decision.



1. Dataset

Use the TSLA daily OHLCV dataset from the course GitHub repository below (a single CSV file, already cleaned — no need to unzip):

https://github.com/kaopanboonyuen/CP020003_ArtificialIntelligence_2026s1/blob/main/dataset/TSLA-KKU-INCLASS.csv

The CSV has one row per trading day, with the following columns:

Column	Meaning
Date	Trading date (YYYY-MM-DD). Sort ascending before doing anything else — never shuffle a time series.
Open / High / Low	Intraday price range for that trading day, in USD.
Close	Raw closing price, not adjusted for stock splits.

Column	Meaning
Adj Close	Closing price adjusted for stock splits (and dividends, where applicable). Tesla split 5-for-1 in August 2020 and 3-for-1 in August 2022 — plot both Close and Adj Close and you will see the raw Close series has two large artificial “cliffs” on those dates that Adj Close does not have.
Volume	Shares traded that day.

Tip: decide up front whether you are forecasting Close or Adj Close, and say why. Forecasting raw Close across a split date without accounting for it will make your model “learn” a fake overnight price collapse.

2. Your Task 🎯

Suggested Investigation Path (guided warm-up)

Before your open-ended investigation, work through these five guided questions from the in-class notebook — they check that your mechanics (chronological split, ADF/ACF/PACF, lag features, sequence windows, scoring) are solid before you go explore on your own:

1. Change TEST_SIZE. Try 30, 90, and 120 trading days instead of 60. How does the naive baseline's RMSE change as the horizon grows? Plot RMSE vs. TEST_SIZE for the naive baseline to show the effect directly.
2. Re-run the ADF test from the in-class notebook on TSLA's Adj Close, then again on the first difference. Report both p-values and state, in one sentence, why a trending stock price is expected to fail the ADF test on the raw series.
3. Plot ACF and PACF on the differenced Adj Close (as in Section 5 of class) and propose an AR/MA order from the plots alone, before fitting anything. Then fit ARIMA with that order and compare it against your first guess.
4. Tune SEQ_LEN for the LSTM and GRU models. Try 10, 20, and 40 days. Does a longer window help, hurt, or make little difference for TSLA? Give a one-sentence hypothesis for why, referencing how volatile TSLA is compared to a broad index like NASDAQ.
5. (Challenge) Forecast across one of the two split dates (August 2020 or August 2022) using raw Close with no adjustment, and compare that forecast's error to the same model run on Adj Close over the same window. Explain, in your own words, why this is a data-cleaning problem and not a modelling problem.

Open-Ended Investigation (the main graded task)

This is where most of your grade comes from. Design a forecasting study around TSLA using the same ladder as class — classical, tabular ML, and deep learning — but with choices that are your own:

- Pick your forecast target and horizon. Close or Adj Close? A 30-day, 60-day, or 90-day held-out test window? Justify the choice.
- Justify your feature set for the ML/DL sections. Which lags, rolling windows, or engineered features (e.g. Volume, High-Low range, day-of-week) did you include, and why do you believe each one carries signal rather than noise?

- Decide what “worth deploying” means for your investigation. Lowest RMSE? Beats naive by the widest margin? Most honest uncertainty interval? Fastest to retrain daily? State your criterion up front, before you start comparing models.
- This is the same kind of design decision our Week 8 assignment asked you to make when choosing a rule-selection criterion for market-basket analysis — here, the design decision is which forecasting objective matters most for a volatile single stock versus a broad index. Document your reasoning.

Required Steps

(mirrors our in-class notebook)

1. EDA & cleaning — load the CSV, parse Date, sort chronologically, and plot Close, Adj Close, and Volume over the full history. Mark the two stock-split dates on your plot and confirm visually that Adj Close removes the cliffs that Close shows.
2. Stationarity & decomposition — run `seasonal_decompose` and the ADF test on your chosen target (raw and differenced), exactly as in Sections 4–5 of class. State whether/how much differencing you need.
3. Chronological train/test split — hold out your chosen number of trading days as a strict future-only test window. State clearly which target column, date range, and `TEST_SIZE` you chose and why.
4. Pick and justify ≥ 2 model families from: Classical (AR/MA/ARMA/ARIMA/SARIMA/Holt-Winters), Tabular ML (lag-feature regressors), and Deep Learning (MLP/RNN/LSTM/GRU/1D CNN). Briefly explain, referencing the recap table from class, why each family is a reasonable fit for a single volatile stock like TSLA.
5. Fit and forecast with each chosen family, on the same train/test boundary and (where applicable) the same feature-engineering rules — no feature may see data from $t+1$ or later (avoid leakage: always `shift(1)` before rolling).
6. Score every model against the naive persistence baseline (“tomorrow = today”) using MAE, RMSE, and MAPE, and note whether each model actually beats naive — for a single stock this is a real possibility, not a formality.
7. Residual diagnostics — for at least one classical model, run `plot_diagnostics()` and a Ljung-Box test; state whether the residuals look like white noise.
8. Handle the stock-split anomaly explicitly — show what happens to at least one model's forecast if it is trained across a split boundary on unadjusted Close, and explain the fix.
9. Business interpretation — in plain language, state which model or family you would trust for TSLA and why, backed by your actual MAE/RMSE/MAPE numbers and how far the forecast horizon is from “just guess yesterday's price.”
10. Conclusion — which family won for your chosen horizon, what trade-off did you accept to get there (e.g. interpretable coefficients vs. raw accuracy, or training time vs. forecast quality), and which model would you actually put in front of a trader first, and why?

Use this table as a starting skeleton for your final scoreboard

(feel free to add rows/columns your investigation needs):

Model	Family	MAE	RMSE	MAPE (%)	Beats naive?
Naive (persistence)	Baseline				—
	Classical				
	ML (tabular)				
	Deep Learning				

⚠ Grading Emphasis — Read This Carefully

Getting one model to run is necessary, but it is not sufficient to score well this week. Most of the marks for this assignment are earned in Steps 6–10 above: how rigorously you benchmark your ≥ 2 model families against the naive baseline on the same test window, and how clearly you can explain, in plain language backed by your own MAE/RMSE/MAPE numbers, which model you would actually trust for TSLA.

A submission that fits one classical model and one ML model, but checks residuals carefully, handles the split anomaly correctly, and backs its recommendation with real numbers, will score higher than a submission that runs five architectures but only offers a single vague sentence of comparison at the end.

3. Bonus Points 🌟

- Build a third model family beyond your required two (e.g. add a deep learning model if your two were Classical + ML) and extend your scoreboard — +bonus
- Add the attention demo from Section 9 of class: compute causal self-attention weights over your last SEQ_LEN days of TSLA and identify which past days the mechanism weights most heavily going into your forecast — +bonus
- Run a VAR model on TSLA alongside a correlated index or peer stock from the course dataset (e.g. NASDAQ/IXIC) and discuss whether the pair forecasts better together than TSLA alone — +bonus
- Quantify the split-handling gap: forecast one window with naive split-blind Close and the same window with Adj Close, and report exactly how much RMSE the split artifact costs you — +bonus
- Combine forecasting with a simple decision rule: if your best model predicts the price will rise by more than X% over the horizon, would a simple long/flat strategy have beaten buy-and-hold over your test window? Show the numbers, with clearly labeled assumptions — +bonus